

Guideline Title: Hypertensive Emergency

Guideline Number: GUID-3203-M010

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Revision dates: 4/30/24	Developed by: Air Care Team
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Signature: 	Signature:
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I. **PURPOSE:** Hypertensive emergency is defined as an elevated blood pressure associated with acute central nervous system, cardiac, and/or renal dysfunction. The purpose is to first recognize the patient with a hypertensive emergency then provide and maintain adequate oxygenation, ventilation, and end-organ tissue perfusion and to provide adequate blood pressure control and expeditious transport to the nearest appropriate medical facility.

II. **DEFINITIONS:**

- a. HR – heart rate
- b. MAP - mean arterial pressure
- c. SBP – systolic blood pressure
- d. DBP – diastolic blood pressure

III. **DEPARTMENT GUIDELINE:**

- a. Establish care as defined by GUID3203-G002 – *General Patient Care*.
- b. Assess patient for headache, visual changes, weakness, chest pain, palpitations, dyspnea, pedal edema, anuria, hematuria, seizures, and encephalopathy. Complete a focused neurologic and cardiovascular exam to include bilateral upper extremity blood pressures.
- c. Isolated asymptomatic hypertension with no evidence of end-organ damage does not require treatment. Monitor closely for changes.
- d. Consider obtaining a 12 lead EKG per GUID3203-PR001 – *12-Lead Electrocardiogram*.
- e. Consider pain medication per GUID3203-G005 *Analgesia and Sedation Management* and apply ETCO2 monitoring.
- f. For BP > 220/120 mmHg with signs of end-organ damage, administer one of the following anti-hypertensives until either symptoms are alleviated or a 20-25% reduction in MAP is achieved.
 - i. **Labetalol 10-20 mg IV** over 2 minutes. May repeat every 10 minutes until goal blood pressure obtained or maximum total dose of 300mg. Avoid if HR < 60 bpm.
 - ii. **Nicardipine infusion at 5mg/h.** Increase by 2.5mg/h IV every 5-15 minutes to maximum of 15mg/h.
 - iii. **Hydralazine 10-20 mg IV** q 20 minutes. Max total dose 40mg. Avoid in elevated intracranial pressure or myocardial infarctions.

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- iv. **Nitroprusside (50 mg/250ml D5W) infusion at 0.3 mcg/kg/min.** The infusion can be increased by 0.5 mcg/kg/min every 5 minutes until desired BP is reached. Max dose 10 mcg/kg/min (Avoid in renal failure).
- v. **Esmolol bolus 500 mcg/kg/min IV** over 1 minute followed by **Esmolol infusion at 50 mcg/kg/min.** Titrate every 5 minutes by repeating 500 mcg/kg/min bolus over 1 minute and then increasing infusion by 50 mcg/kg/min. Maximum dose 300 mcg/kg/min
- g. See appropriate guidelines for specific blood pressure control parameters for these conditions
 - i. Stroke/Transient Ischemic Attack *GUID3203-M017 – Stroke or Transient Ischemic Attack*
 - ii. Subarachnoid Hemorrhage, Non-traumatic per *GUID3203-M013 – Subarachnoid Hemorrhage*
 - iii. Thoracic Aortic Dissection per *GUID3203-M018 – Thoracic Aortic Dissection*
 - iv. Abdominal Aortic Aneurysm (rupture or impending rupture) per *GUID3203-M001 – Abdominal Aortic Aneurysm*
 - v. Congestive Heart Failure & Pulmonary Edema, per *GUID3203-C005 – Congestive Heart Failure and Pulmonary Edema*
- h. Avoid D5W or dextrose-based fluids.
- i. If patient is hypertensive due to sympathomimetic drug use (e.g., cocaine, methamphetamines, bath salts, ecstasy, etc.) treat with benzodiazepines and AVOID beta-blockers
 - i. **Midazolam 0.05-0.1 mg/kg IV/IO**, may repeat every 3-5 minutes PRN, maximum dose 5 mg
- j. Continuously monitor heart rate and blood pressure during all medication administration.
- k. Consider obtaining atrial line placement if at sending facility and continue atrial blood pressure monitoring for patients being transported on antihypertensive medications
- IV. **DOCUMENTATION:**
 - a. EMS Charts
- V. **REFERENCES:**